One Step Function Explains Why Galaxies Die

Gas supply versus gas depletion timescales unify black hole and halo quenching across 350,000 galaxies.

arXiv:2610.20505 — Step-like quenching from gas mass conservation

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Background

Galaxies stop forming stars through many routes, from black hole feedback to halo heating. A single framework tying these mechanisms together has remained elusive.

> **No simple, unified rule links diverse quenching processes to observable galaxy properties.**

What this paper is selling

Gas mass conservation yields a step-function quenching response set by supply versus depletion timescales. Fitting it to quenched fractions infers the gas-supply timescale directly from data.

Key results to remember

- Characteristic gas-supply timescale is about 1 Gyr across the sample.
- Black holes drive inside-out quenching; dark matter halos drive outside-in quenching.
- Intermediate surface density galaxies sit at a critical quenching-rejuvenation state.

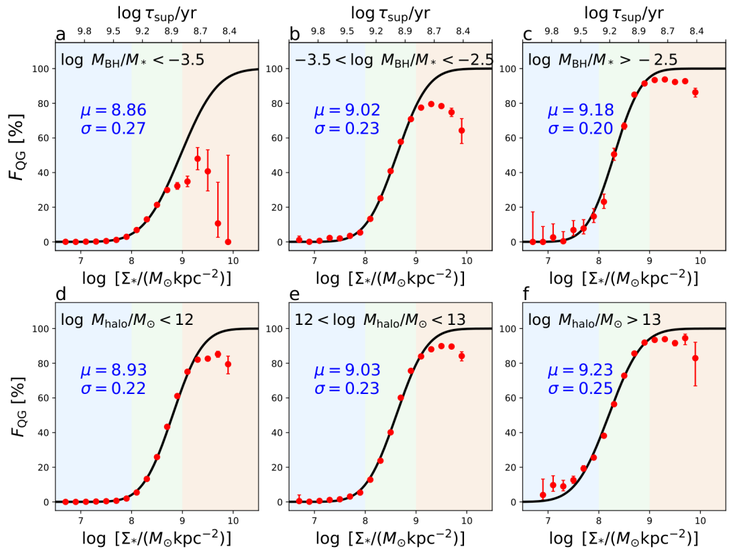


Figure 2 | Black hole growth lengthens and narrows supply timescales, while halo mass lengthens and broadens them.

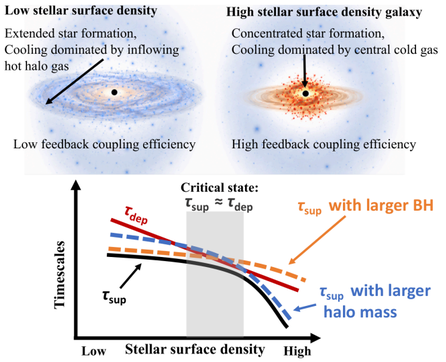


Figure 4 | At intermediate surface density, supply and depletion timescales converge, making quenching most sensitive to regulation.